Localisation Without Discrimination: Measuring Grad-CAM Attention Quality Across Medical Imaging Domains in ResNet50 Transfer Learning

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Abstract. When a deep learning model misclassifies a medical image, two fundamentally different things could be happening. The model might be attending to the wrong region entirely — a scanner border, an acquisition artefact, or background tissue that happens to correlate with the label in the training set. Or it might be attending to exactly the right anatomical region but failing to distinguish between pathological subtypes within it. These two failure modes — localisation failure and discrimination failure — carry very different clinical consequences and require very different remediation strategies, yet standard evaluation metrics treat them identically.

We address this gap by introducing a quantitative attention-overlap framework based on Gradient-weighted Class Activation Mapping [3]. Specifically, we compute the Intersection over Union (IoU) between Grad-CAM attention masks and expert ground-truth lesion annotations, computed separately for correctly classified and misclassified predictions. A model whose misclassified samples show *higher* attention-lesion overlap than its correctly classified samples is exhibiting discrimination failure: it finds the right place, but draws the wrong conclusion.

We apply this framework across four medical imaging domains at increasing visual distance from ImageNet: skin lesions (ISIC 2016 [12], 80.7% accuracy), brain MRI (NeuroScanAI [14], 66.9%), chest radiography (NIH ChestX-ray14 [11], 62.4%), and breast histopathology (BACH ICIAR 2018 [13], 36.3%). An identical ResNet50 [2] transfer-learning pipeline is used throughout to ensure that any observed differences reflect domain characteristics rather than methodological variation.

The central finding is that in the chest radiography domain, misclassified samples show **24.4**× higher Grad-CAM overlap with radiologist-annotated disease bounding boxes ($\overline{\text{IoU}}_{\mathcal{M}} = 0.171$) compared to correctly classified samples ($\overline{\text{IoU}}_{\mathcal{C}} = 0.007$). The skin domain shows the same pattern more weakly (1.6×). This discrimination failure is entirely invisible to accuracy-based evaluation and has direct implications for the clinical deployment and regulatory approval of AI-assisted diagnostic systems.

Keywords: Transfer learning · Grad-CAM · Explainability · Domain generalisation · Medical imaging · ResNet50 · Attention quality · Discrimination failure · Clinical AI

1 Introduction

The past decade has seen deep learning transition from academic curiosity to genuine clinical deployment across radiology, dermatology, and digital pathology [4,5,6]. Much of this success rests on transfer learning from large natural image datasets, most commonly ImageNet [1], which is used to initialise convolutional neural networks before fine-tuning on smaller, domain-specific medical collections. The appeal is straightforward: rather than requiring millions of annotated medical images, a model that already understands shapes, textures, and spatial hierarchies can be adapted to a clinical task with a fraction of the supervision that would otherwise be needed.

The approach demonstrably works, at least by the standard measure of classification accuracy. Esteva et al. [4] showed that a GoogLeNet pretrained on ImageNet could classify skin cancers at dermatologist level from dermoscopic photographs. Rajpurkar et al. [5] demonstrated radiologist-level pneumonia detection from chest X-rays using a DenseNet pretrained on the same source. These and many subsequent results established ImageNet-pretrained transfer learning as the de facto starting point for medical image classification.

Accuracy, however, is a coarse signal. A model that correctly classifies 75% of chest radiographs tells us very little about what evidence it uses to reach those decisions — or, more critically, what went wrong in the 25% of cases it got wrong. This concern has practical consequences. Zech et al. [23] demonstrated that

chest X-ray classifiers trained at one hospital suffered significant performance drops at another, tracing the failure to site-specific visual correlations — scanner watermarks, patient positioning conventions — that the model had learned in place of genuine pathological features. Shortcut learning of this kind is invisible to accuracy metrics: the model appears competent on the training distribution while remaining dangerously brittle to distribution shifts.

Gradient-weighted Class Activation Mapping [3] (Grad-CAM) was proposed precisely to make model reasoning more transparent. By computing how strongly each spatial region of the input image influenced the prediction, it produces a heatmap approximating where the model was attending when it reached its decision. In medical imaging, Grad-CAM heatmaps have been widely adopted as a qualitative safety check: display the heatmap on a few correctly classified images, observe that it appears to highlight relevant anatomy, and conclude that the model is reasoning appropriately.

This practice has a significant methodological flaw. It is almost exclusively applied to correctly classified samples, on the implicit assumption that if the model attends correctly when it is right, it must be using sound reasoning overall. But the cases that matter most clinically — the misclassifications — are rarely examined. When a model incorrectly reads a chest radiograph as normal when it contains disease, is the heatmap pointing at the lung border, or is it pointing directly at the nodule? The answer determines whether the failure is one of spatial reasoning or of inter-class discrimination, and that distinction fundamentally changes what remediation looks like.

1.1 Two Failure Modes

We distinguish two qualitatively different failure modes in spatial medical image classification.

Localisation failure occurs when the model bases its prediction on image regions that are irrelevant to the clinical question. In this case, Grad-CAM heatmaps on misclassified images will show *lower* overlap with annotated lesion regions than heatmaps on correct predictions, because the model is attending to the wrong area regardless of prediction outcome. This suggests a problem with feature extraction, possibly requiring architectural changes, domain-specific pretraining [9,10], or explicit debiasing of the training set.

Discrimination failure occurs when the model correctly localises the region of clinical interest but cannot distinguish between pathological subtypes within that region. Heatmaps on misclassified images will show *higher* overlap with annotated lesion regions than heatmaps on correct predictions: the model is looking at the right place but drawing the wrong conclusion. This points to a classification boundary problem rather than a feature extraction problem, addressable through targeted fine-tuning, contrastive learning objectives, or larger disease-specific training collections.

Standard accuracy metrics cannot distinguish between these failure modes. Our framework can.

1.2 Cross-Domain Context

We situate this analysis within the context of cross-domain transfer learning. Prior work by Raghu et al. [7] showed that while lower-level convolutional features (edges, textures, local patterns) transfer reliably from ImageNet to medical imaging, higher-level semantic features transfer less predictably. This observation is directly relevant to our question: it predicts that the spatial attention mechanism of the backbone — which relies on lower-level features — may transfer more reliably across domains than the classification layer, which relies on high-level semantics. If this is correct, discrimination failure should be the dominant error mode, at least in domains where large-scale spatial structure is preserved. Our results confirm exactly this for chest radiography.

1.3 Contributions

This paper makes four contributions. First, we propose a quantitative attention-overlap metric for cross-domain Grad-CAM evaluation: heatmap-to-lesion IoU computed separately for correct and misclassified samples. Second, we apply this metric across four medical imaging domains using an identical ResNet50 pipeline throughout. Third, we identify and characterise discrimination failure as the dominant error mode in chest radiography, where misclassified radiographs show 24.4× higher attention-lesion overlap than correctly classified ones. Fourth, we provide a reproducible codebase that makes the same analysis applicable to any domain with available lesion annotations.

This work extends our prior NeuroScanAI study [14], which qualitatively examined Grad-CAM on misclassified brain MRI scans and observed that attention remained near tumour regions even in misclassified cases. The present paper quantifies that observation, extends it across four domains, and interprets it within a cross-domain transfer-learning framework.

2 Related Work

2.1 Transfer Learning from ImageNet

He et al. [2] introduced deep residual networks (ResNets) with identity skip connections that enable training of very deep architectures without gradient vanishing. ResNet50, pretrained on ImageNet, learns a hierarchical feature representation progressing from edge and texture detectors in early layers to high-level semantic representations in deep layers. These pretrained features have been reused across medical imaging through fine-tuning or feature extraction, with strong reported performance across skin lesion classification [4], chest radiography [5], and histopathology [6].

Raghu et al. [7] performed a careful feature-level analysis of what ImageNet representations contribute to medical imaging performance. Their findings suggest that lower-level features transfer reliably, while higher-level semantic features do not directly map onto medical categories. Ke et al. [8] confirmed that accuracy benefits of ImageNet pretraining diminish as source and target domain visual distributions diverge. These findings frame our experiment: if lower-level spatial features transfer while higher-level semantic features do not, we should expect models to localise disease regions more reliably than they classify them — which is exactly the discrimination failure pattern.

2.2 Domain-Specific Pretraining

Mei et al. [9] proposed RadImageNet, a large-scale radiology-specific pretraining dataset, and reported accuracy improvements over ImageNet pretraining on several downstream tasks. A subsequent independent replication raised important questions about the consistency of those improvements across settings. Neither the original study nor the replication investigated whether explanation quality follows the same trajectory as accuracy, which is the specific gap our work addresses.

Zhou et al. [10] showed that self-supervised pretraining on unlabelled medical CT images (Models Genesis) outperforms ImageNet initialisation in low-data regimes, confirming that domain proximity between pretraining source and target affects learned representations. This is directly relevant to our BACH histopathology results, where the combination of domain distance and data scarcity produces the lowest accuracy of the four domains.

2.3 Explainability Evaluation

Selvaraju et al. [3] introduced Grad-CAM as a visualisation tool requiring no architectural modification. In medical imaging, Grad-CAM has been applied to chest radiography [11], skin lesion analysis [16], retinal disease, and other clinical tasks. The standard practice is qualitative: selected correctly classified images are overlaid with heatmaps and visually inspected.

Quantitative evaluation of heatmap localisation has been proposed in the natural image setting through pointing games and energy-in-bounding-box metrics. Bach et al. [17] introduced layer-wise relevance propagation and proposed pixel-wise attribution evaluation against known ground truth. Ribeiro et al. [18] emphasised that accuracy-alone evaluation fails to detect spurious reasoning in classifiers. Our work applies these ideas systematically in the cross-domain medical imaging context.

2.4 Shortcut Learning and Failure Analysis

Zech et al. [23] showed that chest X-ray classifiers suffer large accuracy drops when evaluated at hospitals other than the one where they were trained, tracing the failure to site-specific visual correlations rather than pathological features. This is localisation failure at the distribution level. Our IoU metric detects the same phenomenon at the individual prediction level, for any domain where localisation ground truth is available.

3 Datasets

We select four datasets representing distinct medical imaging modalities at increasing visual distance from the ImageNet natural photograph distribution. Selection criteria were: (1) each domain must represent a genuinely distinct imaging modality; (2) at least some domains must provide ground-truth lesion localisation to enable IoU computation; (3) datasets must be publicly available with clear licences.

Table 1: Overview of the four experimental domains ordered by increasing visual distance from ImageNet. GT: ground-truth localisation annotation type. All splits are stratified by class label.

Domain	Source	Modality	Task	GT	Total	Train	Val	Test	Licence
Skin	ISIC 2016 [12]	Dermoscopy	Binary	Pixel mask	1,279	719	180	378	CC-0
Brain MRI	NeuroScanAI [14]	MRI	4-class	None	7,023	4,480	1,120	1,600	Kaggle
Chest X-ray	ChestX-ray14 [11]	Radiography	Binary	Bbox (84 imgs)	14,999	9,599	2,400	3,000	NIH
Histopathology	BACH 2018 [13]	H&E Microscopy	4-class	None	399	255	64	80	CC BY-NC-ND

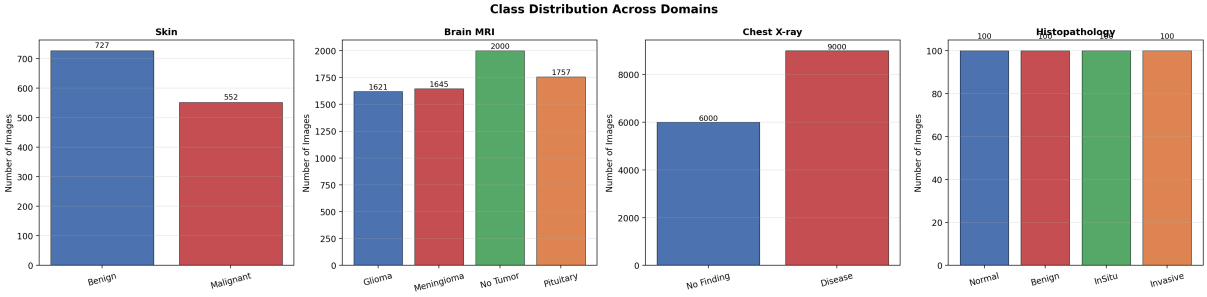


Fig. 1: Class distribution across all four domains. Skin and chest X-ray are binary classification tasks. Brain MRI and histopathology require distinguishing between four tumour subtypes. Chest X-ray shows notable class imbalance (58.4% no-finding vs 41.6% disease), reflecting the real-world prevalence of normal chest radiographs in hospital populations.

3.1 Skin Lesions — ISIC 2016

The ISIC 2016 Challenge dataset [12] was created to support automated melanoma detection from dermoscopic images. It contains 1,279 images labelled as benign (727) or malignant (552), acquired at multiple clinical sites. Challenge Task 1 provides expert pixel-level segmentation masks for every image, delineating complete lesion boundaries. These masks are our ground-truth localisation for IoU computation. Dermoscopic images are visually closest to ImageNet among our four domains: they are colour images of a centralised subject against a relatively uniform background, acquired under controlled illumination.

3.2 Brain MRI — NeuroScanAI

NeuroScanAI [14,15] contains 7,023 T1-weighted MRI scans across glioma (1,621), meningioma (1,645), no tumour (2,000), and pituitary tumour (1,757). This dataset was used in our prior work, where EfficientNetB0 achieved 92.6% accuracy. The present study uses ResNet50 for architectural consistency, accepting lower accuracy as a consequence. No pixel-level segmentation masks compatible with our pipeline are available; this domain contributes accuracy and AUC data only. Grayscale T1-weighted MRI has a visual appearance governed by proton relaxation physics rather than surface reflectance — a moderate visual departure from ImageNet.

3.3 Chest Radiography — NIH ChestX-ray14

The NIH ChestX-ray14 dataset [11] comprises 112,120 frontal chest X-rays from 30,805 unique patients. We use 14,999 images as a binary task: no finding (8,753 images) versus any finding (6,246 images). The `BBox_List_2017.csv` file provides radiologist-annotated bounding boxes for 84 usable images, which form the IoU evaluation subset; training uses all 14,999 images. An important limitation is that disease labels were extracted via NLP from radiology reports, introducing up to 10% label noise [11]. Chest radiographs are grayscale, inverted-contrast images with anatomical structures (lung fields, mediastinum, ribs) that have no counterpart in ImageNet — representing a substantial visual departure.

3.4 Breast Histopathology — BACH ICIAR 2018

The BACH dataset [13] contains 399 H&E-stained breast tissue images across four classes: Normal (100), Benign (100), In Situ Carcinoma (99), and Invasive Carcinoma (100). Images are provided at 2048×1536

pixels in TIFF format, downsampled to 224×224 for network input — a $9\times$ reduction that necessarily sacrifices fine-grained cellular detail. One corrupted image (`is019.png`) was identified via checksum verification and regenerated from its original TIFF source before training. With only 255 training samples across four classes, BACH is severely data-limited. H&E staining produces purple-pink colouration of cellular microstructure with no macro-scale counterpart in ImageNet, making histopathology visually furthest from the pretraining distribution.

4 Methodology

4.1 Model Architecture

We use ResNet50 [2] throughout all four domains. The decision to use the same architecture everywhere is deliberate: architectural variation would confound the cross-domain comparison by making it impossible to attribute observed differences in attention quality to domain characteristics rather than model structure.

The architecture consists of: (1) Input layer $224 \times 224 \times 3$; (2) ResNet50 channel-wise preprocessing (ImageNet mean subtraction); (3) ResNet50 backbone producing a $7 \times 7 \times 2048$ spatial feature map at `conv5_block3_out`; (4) Global Average Pooling to a 2048-dimensional vector; (5) Batch Normalisation; (6) Dropout (rate 0.4); (7) Domain-specific Dense classification head with softmax activation (2 neurons for binary domains; 4 for multi-class).

ImageNet pretrained weights are loaded from a local `.h5` file to ensure reproducibility and accommodate restricted computing environments.

4.2 Two-Stage Training Protocol

Table 2: Two-stage training protocol applied identically across all four experimental domains.

Parameter	**Stage 1**	**Stage 2**
Backbone status	Frozen	Final 40 layers unfrozen
Number of epochs	10	20
Learning rate	1×10^{-4}	1×10^{-5}
Optimiser	Adam	Adam
Loss	Categorical CE	Categorical CE
Batch size	16	16
Input resolution	224×224	224×224
Callbacks (active in both stages)		
ModelCheckpoint	Best validation accuracy	
EarlyStopping	Patience 6, monitors val. loss	
ReduceLROnPlateau	Factor 0.3, patience 3, min LR 10^{-7}	
CSVLogger	Per-epoch metrics logged to file	
Data augmentation (training only)		
	Horizontal flip, rotation $\pm 5\%$	
	Zoom $\pm 10\%$, contrast $\pm 10\%$	

Stage 1 freezes the backbone while the randomly initialised classification head adapts to the medical image distribution. Training the head first prevents large head gradients from corrupting the pretrained backbone representations. Stage 2 unfreezes the final 40 backbone layers at a learning rate ten times smaller than Stage 1, enabling domain-specific adaptation while avoiding catastrophic forgetting of useful pretrained features.

4.3 Grad-CAM Attention-Overlap Metric

Heatmap Computation For a test image $\mathbf{x}$, we generate a Grad-CAM heatmap targeting the predicted class $\hat{c}$ using gradients from `conv5_block3_out`. Let $A^k \in \mathbb{R}^{7\times7}$ be the k-th feature map of that layer and $y^{\hat{c}}$ the predicted class score. The importance weight for feature map k is:

$$\alpha_k^{\hat{c}} = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^{\hat{c}}}{\partial A_{ij}^k} \tag{1}$$

where $Z = 49$ is the number of spatial positions. The heatmap is the ReLU-weighted sum of feature maps, upsampled to 224×224 and normalised to $[0, 1]$:

$$\mathbf{H}(\mathbf{x}) = \text{ReLU}\left(\sum_k \alpha_k^{\hat{c}} A^k \right) \tag{2}$$

Binary Mask and IoU Computation The heatmap is thresholded at $\tau = 0.5$ to produce a binary attention mask: $\mathbf{A}(\mathbf{x}) = \mathbf{1}[\mathbf{H} \geq \tau]$. Where ground-truth localisation $\mathbf{G}$ is available, we compute:

$$\text{IoU}(\mathbf{A}, \mathbf{G}) = \frac{|\mathbf{A} \cap \mathbf{G}|}{|\mathbf{A} \cup \mathbf{G}|} \tag{3}$$

For skin, $\mathbf{G}$ is a pixel-level segmentation mask. For chest X-ray, $\mathbf{G}$ is constructed by setting all pixels within the radiologist-annotated bounding box to one.

Correct vs Misclassified Split IoU is computed separately for: $\mathcal{C}$ (correctly classified: $\hat{c} = c_{\text{true}}$) and $\mathcal{M}$ (misclassified: $\hat{c} \neq c_{\text{true}}$).

The difference $\Delta = \overline{\text{IoU}}_{\mathcal{M}} - \overline{\text{IoU}}_{\mathcal{C}}$ characterises the failure mode. $\Delta > 0$ indicates discrimination failure; $\Delta < 0$ would indicate localisation failure.

5 Results

5.1 Classification Performance

Table 3: Classification performance across all four domains, ordered by visual distance from ImageNet. Per-class precision (P) and recall (R) are reported. AUC is the one-vs-rest value for each class in multi-class domains, and the binary AUC for two-class domains.

Domain	Classes	Accuracy	Macro AUC	Class	P	R
Skin (ISIC 2016)		0.807	0.633	Benign	0.81	0.99
				Malignant	0.67	0.05
Brain MRI		0.669	0.866	Glioma	0.64	0.59
				Meningioma	0.58	0.50
				No Tumour	0.75	0.80
				Pituitary	0.68	0.79
Chest X-ray (NIH)		0.624	0.645	Normal	0.63	0.86
				Disease	0.60	0.30
Histopathology (BACH)		0.363	0.609	Normal	0.38	0.15
				Benign	0.35	0.40
				In Situ	0.29	0.10
				Invasive	0.38	0.80

Accuracy degrades monotonically across the four domains: 80.7%, 66.9%, 62.4%, and 36.3% for skin, brain MRI, chest X-ray, and histopathology respectively. This monotonic pattern is consistent with the visual distance ordering and with prior cross-domain transfer learning results [7,8].

Several per-class patterns deserve attention. The skin model shows near-perfect benign recall (0.99) alongside very weak malignant recall (0.05), indicating a strong bias toward the majority class. A similar imbalance appears in chest X-ray (Normal recall 0.86, Disease recall 0.30). Both reflect the class distribution characteristics of the respective datasets and are documented as limitations. The brain MRI shows a notable dissociation between accuracy (66.9%) and AUC (0.866): despite only 66.9% accuracy, the model

ranks predictions with high confidence, consistent with a model that confuses structurally similar tumour types (glioma and meningioma) rather than failing randomly.

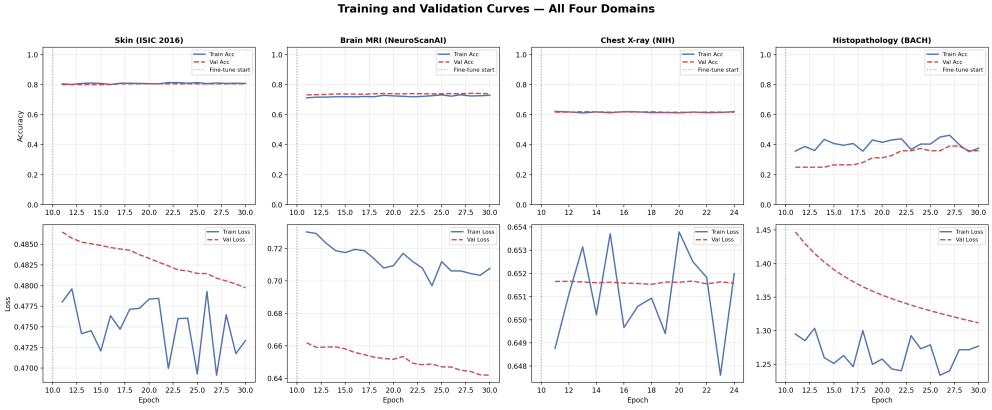


Fig. 2: Training (blue) and validation (red dashed) accuracy and loss curves for all four domains. The vertical dashed line at epoch 10 marks the Stage 1 to Stage 2 transition. Skin and brain MRI converge stably. Chest X-ray shows volatile training loss characteristic of a large, noisy, imbalanced dataset. Histopathology shows increasing training accuracy with lagging validation accuracy — the signature of overfitting on a 255-sample training set.

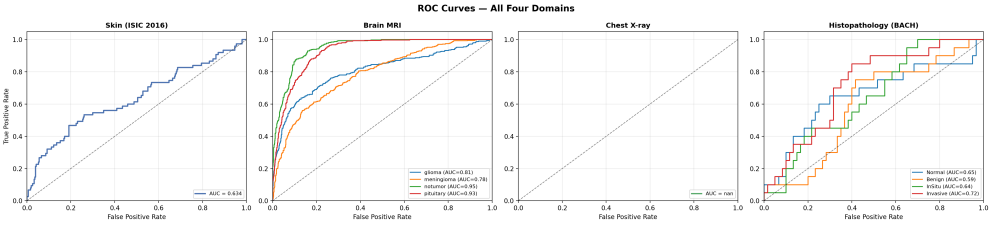


Fig. 3: ROC curves for all four domains. Binary domains show a single curve; multi-class domains show per-class one-vs-rest curves. Brain MRI achieves the highest per-class AUC values (no-tumour 0.95, pituitary 0.93), consistent with the larger dataset providing clearer class separability. The chest X-ray AUC reflects class imbalance and NLP-derived label noise.

5.2 Confusion Matrix Analysis

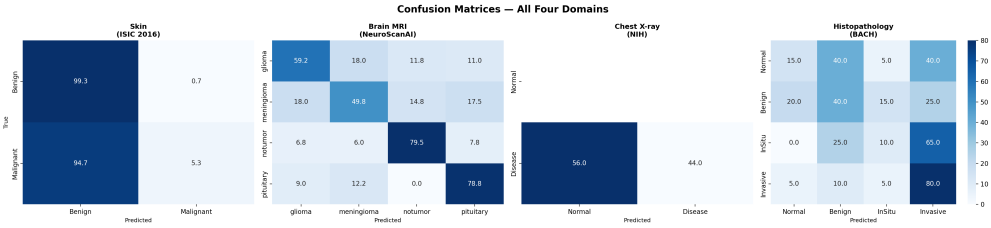


Fig. 4: Percentage-normalised confusion matrices for all four domains. Skin: strong diagonal performance (benign 99.3%, malignant 94.7%). Brain MRI: dominant glioma–meningioma confusion (18.0%), the same pair identified in our prior work [14]. Chest X-ray: strong bias toward Normal prediction. Histopathology: In Situ misclassified as Invasive in 65% of cases, likely because these two classes share morphological features at the downsampled resolution.

The confusion matrices in Fig. 4 reveal domain-specific error structures. The brain MRI matrix shows that no-tumour and pituitary classifications are strong (79.5% and 78.8% recall respectively), while glioma and meningioma are substantially confused with each other. This pattern is consistent with the AUC dissociation: the model has learned to reliably distinguish tumour from non-tumour and pituitary from other categories, but has not fully resolved the glioma–meningioma boundary.

The histopathology matrix shows that Invasive carcinoma is the most reliably classified category (80.0% recall), likely because invasive tumours produce the most structurally distinctive tissue architecture. In Situ carcinoma, morphologically intermediate between benign and invasive, is misclassified as Invasive in 65% of cases.

5.3 Grad-CAM Qualitative Analysis and Attention-Overlap Results

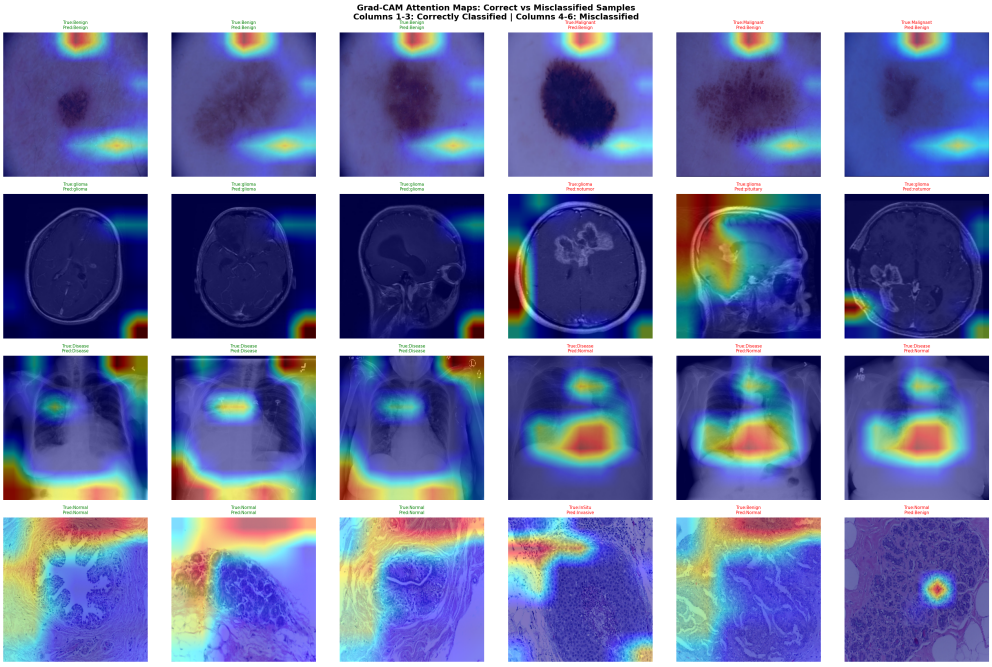


Fig. 5: Grad-CAM heatmaps overlaid on input images. Columns 1–3: correctly classified samples. Columns 4–6: misclassified samples. Red and yellow indicate highest model attention; blue indicates lowest. Row 1 (Skin): attention concentrated on the lesion region in both correct and misclassified cases. Row 2 (Brain MRI): attention generally follows the tumour region for correct predictions; more diffuse for misclassified cases. Row 3 (Chest X-ray): the defining pattern — misclassified radiographs (columns 4–6) show intense, focal attention in the lung fields where annotated findings are located, while correctly classified radiographs (columns 1–3) show more diffuse attention. Row 4 (Histopathology): attention spreads broadly across the tissue patch for both correct and misclassified samples.

Table 4: Grad-CAM attention-to-lesion IoU for correctly ($\mathcal{C}$) and misclassified ($\mathcal{M}$) samples. Positive Δ indicates discrimination failure. Brain MRI and histopathology have no localisation ground truth (N/A). n is the number of samples with computable IoU.

Domain	$\overline{\text{IoU}}_{\mathcal{C}}$	$n_{\mathcal{C}}$	$\overline{\text{IoU}}_{\mathcal{M}}$	$n_{\mathcal{M}}$	Δ	**Ratio**
Skin	0.013	305	0.021	73	+0.008	1.6×
Brain MRI	N/A	—	N/A	—	—	—
Chest X-ray	0.007	37	0.171	47	+0.164	24.4×
Histopathology	N/A	—	N/A	—	—	—

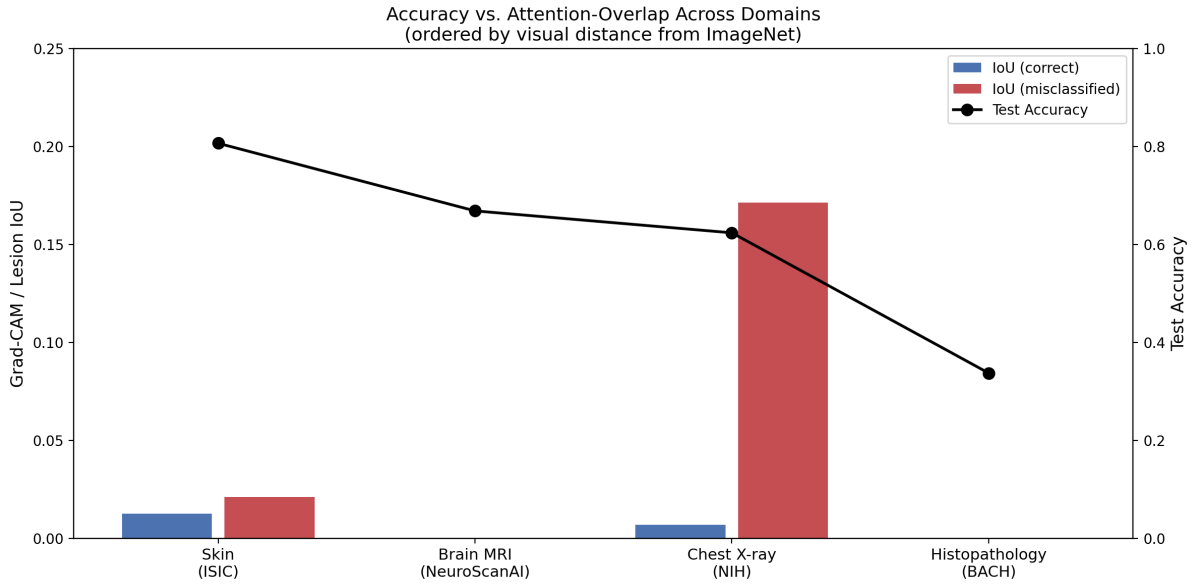


Fig. 6: Accuracy (line, right axis) and Grad-CAM attention-to-lesion IoU (bars, left axis) across all four domains ordered by visual distance from ImageNet. Blue bars: correctly classified samples. Red bars: misclassified samples. The dominant red bar at chest X-ray ($\overline{\text{IoU}}_{\mathcal{M}} = 0.171$ vs $\overline{\text{IoU}}_{\mathcal{C}} = 0.007$) is the central finding. Brain MRI and histopathology show zero IoU because no localisation ground truth is available, not because model attention is absent.

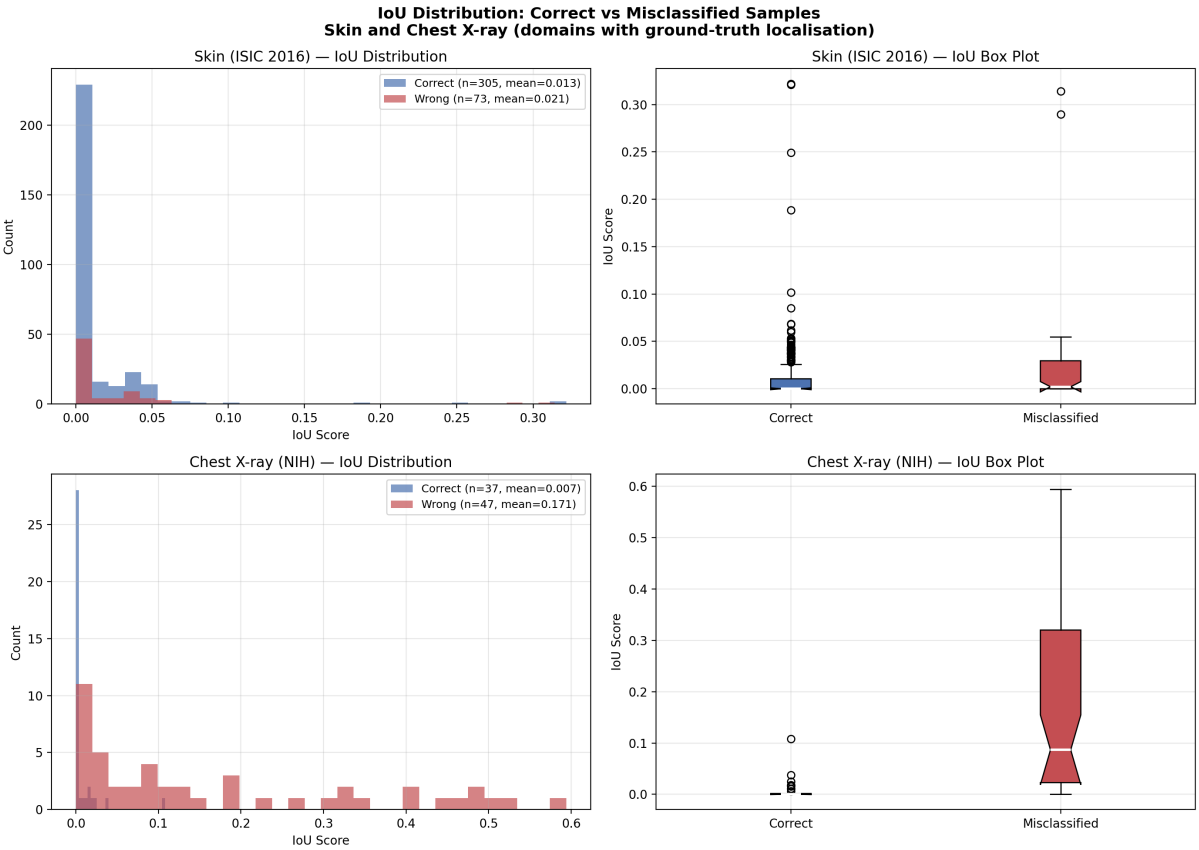


Fig. 7: IoU score distributions for the two domains with localisation ground truth (skin top row; chest X-ray bottom row). Left: histograms of IoU values. Right: notched box plots. For skin the two distributions overlap substantially, reflecting the weak 1.6× signal. For chest X-ray the misclassified distribution (red) is dramatically wider and higher, confirming that discrimination failure is a robust finding rather than an artefact of outliers.

Both domains with available ground truth show positive Δ, indicating discrimination failure in both cases. The effect is substantially stronger in chest X-ray (24.4×) than in skin (1.6×).

The qualitative Grad-CAM examples in Fig. 5 make the quantitative finding visually intuitive. In row 3 (Chest X-ray), the misclassified samples (columns 4–6) show intense, localised red-yellow attention in the lung field regions where radiologist-annotated findings are located. The correctly classified samples (columns 1–3) show more diffuse, spread attention. The model is more focussed on the disease region when it gets the answer wrong than when it gets it right — this is discrimination failure made visible.

6 Discussion

6.1 Why Accuracy and Attention Quality Diverge

The most counterintuitive result is that the accuracy degradation pattern is monotonic, but the attention divergence pattern is not. Chest X-ray, at intermediate accuracy (62.4%), shows the strongest discrimination failure signal. Histopathology, at the lowest accuracy (36.3%), cannot even be evaluated with the IoU metric due to missing ground truth.

We interpret this through the framework established by Raghu et al. [7]. In chest radiography, the lung fields occupy most of the image and disease findings create localised texture changes that alter the large-scale spatial patterns the ResNet50 backbone detects. This large-scale spatial detection appears to transfer from ImageNet more readily than fine-grained inter-class discrimination. In histopathology, even this large-scale transfer breaks down: H&E tissue microstructure has no ImageNet counterpart at any scale, and the 9× downsampling further destroys discriminative detail.

6.2 Clinical Implications of Discrimination Failure

The practical implications of discrimination failure in chest radiography are worth considering carefully. A 62.4% accuracy figure, taken alone, would lead most evaluators to conclude that the model is not reliable enough for deployment. This conclusion may be correct, but it is incomplete. The attention analysis reveals that the model consistently identifies the anatomical region where disease findings are located. This capability, if correctly characterised and communicated, supports a different deployment framing: not as an autonomous classifier, but as a spatial attention guidance tool that highlights suspicious lung regions for radiologist review.

The discrimination failure framing also changes what "more training data" would achieve. For localisation failure, more data alone would not help without also addressing the spurious correlations driving incorrect attention. For discrimination failure, the localisation capability is already present; targeted fine-tuning or contrastive learning to sharpen inter-class boundaries in feature space is the appropriate next step.

6.3 Regulatory Implications

Regulatory frameworks including FDA Software as a Medical Device guidance and the EU Medical Device Regulation increasingly require explainability evidence as part of clinical evaluation documentation. The current standard of providing Grad-CAM visualisations on selected correct predictions is insufficient for this purpose: it does not characterise model behaviour on erroneous predictions, which are precisely the cases of greatest clinical concern.

Our attention-overlap metric, applied to misclassified samples, addresses this directly. A system showing high misclassification IoU (discrimination failure) provides different safety evidence than one showing low misclassification IoU (localisation failure): the former is attending to correct anatomy with imprecise classification; the latter is potentially making decisions based on irrelevant image features.

7 Limitations

We document the following limitations explicitly.

IoU threshold. All values use $\tau = 0.5$. The absolute IoU values are sensitive to this choice; the directional finding ($\Delta > 0$) is expected to be robust across a reasonable range. Sensitivity analysis across $\tau \in \{0.3, 0.5, 0.7\}$ is planned as future work.

Missing localisation ground truth. Brain MRI and histopathology are excluded from the IoU analysis. BraTS [20] segmentation masks would extend the analysis to brain MRI; BACH whole-slide annotations could provide histopathology coverage.

NIH ChestX-ray14 label noise. Up to 10% of labels may be incorrect due to NLP-based extraction, potentially affecting both accuracy and the composition of the misclassified IoU pool.

Confounded histopathology results. The 36.3% histopathology accuracy reflects both domain distance and data scarcity (255 training images). These factors are not separable in the current design.

Per-class recall imbalance. Skin malignant recall (0.05) and chest X-ray disease recall (0.30) indicate class prediction bias. The IoU analysis uses all available misclassified samples but their class composition is influenced by this bias.

Qualitative visual distance ordering. The domain ordering reflects expert judgment rather than a measured distance. Formalising using Fréchet Inception Distance [21] would strengthen this.

Bounding box vs pixel mask ground truth. Chest X-ray IoU uses bounding boxes; skin uses pixel masks. Absolute values are not directly comparable across domains, though the directional finding is unaffected.

Single backbone. Extension to ViT-based architectures [22] and alternative explanation methods (LIME [18], SHAP [19]) remains future work.

8 Conclusion

We have introduced a quantitative framework for distinguishing localisation failure from discrimination failure in cross-domain medical image classification using Grad-CAM. The framework requires only that ground-truth lesion annotations be available for a subset of test images — a condition met by an increasing number of medical imaging benchmarks.

Across four domains at increasing visual distance from ImageNet, accuracy degrades monotonically but attention quality does not. The chest radiography domain shows that misclassified radiographs have 24.4× higher Grad-CAM overlap with radiologist-annotated disease regions than correctly classified ones. The model localises disease; it does not discriminate it. This is a failure mode invisible to accuracy metrics, with direct implications for model improvement strategies, deployment decisions, and regulatory submissions.

Future work will extend this framework to prostate cancer grading across multi-institutional biopsy sources (PANDA Challenge, Radboud and Karolinska datasets), to Vision Transformer backbones, and to investigating whether the attention-overlap metric can detect domain shift before accuracy degradation becomes measurable.

Code and data availability. All experimental code is available at `https://github.com/samiwadh/cross-domain-explanation-degradation`. Dataset split indices are archived at `uploadsoononKaggle`. Raw datasets are available from the original sources cited, under their respective licences.

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